

Qualifying Exam in Microeconomics
Arizona State University
August 2016

There are four questions and you have four hours to complete the exam. If you think a question is in error, please point out why clearly why you think it is error and continue working.

1. The two only firms in a given industry must decide their respective advertising expenditures. Profit for firm i , $i = 1, 2$, is

$$\Pi_i(x_i, x_j) = x_i(2 - x_i) - x_j,$$

where x_i and x_j are, respectively, firm i 's and j 's advertising expenditures.

- (a) Suppose both firms choose their expenditures simultaneously and independently. Please find a Nash equilibrium in pure strategies. What are the Nash equilibrium profits?
- (b) Is the solution found in (a) Pareto Optimal? Please compute the Pareto optimal symmetric solution (i.e., both firms obtain the same profit). Why does the solution to (a) fail to be Pareto optimal?
- (c) Suppose now that firms do not act independently. Each firm can guarantee for itself a profit of zero by withdrawing from the market (i.e., the reservation profit is zero). Suppose the firms will only agree on an outcome that is on the core. Please characterize (i.e., describe mathematically) the solutions that are in the core.
2. Consider an economy with n identical people, a single physical good, and a finite number $S > 1$ states of the world. Each person has a riskless initial endowment of $w > 0$ in each state of the world; and has subjective expected utility preferences over state contingent consumption with belief $\pi_s > 0$ that state s occurs, and a C_2 vN-M utility u with $u' > 0$ and $u'' < 0$.

There is a project available that adds $x_s \geq 0$ units of the good to aggregate wealth in state s , with

$$(i) \sum_s \pi_s x_s > 0 \quad \text{and} \quad (ii) \quad u(w) > \sum_s \pi_s u(w + x_s).$$

- (a) Explain in words what (i) and (ii) assert.

Consider the following risk-sharing game. Each of the n people simultaneously and independently says 'yes' or 'no' to the project. Each person that says *no* simply consumes w in each state. If m people say *yes*, then each of them consumes state-contingent wealth

$$(w + \frac{x_1}{m}, \dots, w + \frac{x_S}{m})$$

- (b) Prove that, for every integer n , there is at least one Nash equilibrium of this game.
- (c) Prove that, if n is large enough, there is a Nash equilibrium in which all n people say *yes*. (Hint: mean-value theorem. The notation $\alpha = \frac{1}{n}$ might suggest an approach.)

3. Consider an economy with two goods, x and y , and a consumer with preferences on $\mathbb{R}_+^2$ represented by

$$u(x, y) = -e^{-x} + e^y.$$

The function u is strictly quasiconcave, even though e^y is strictly convex, a fact you should use in what follows. Normalize the price of good y to be 1, let $p > 0$ be the relative price of x in terms of y , and let the consumer's normalized wealth be $w > 0$.

- Write down the Kuhn-Tucker conditions for the utility maximization problem.
- Suppose that the demand for *both* goods is positive at some $(p_0, w_0) \gg 0$. What therefore can you say about monotonicity of the demand for x in w in a neighborhood of w_0 (at $p = p_0$)? We suggest that you consider the cases $p_0 \geq 1$ and $p_0 < 1$ separately.
- What properties of u account for the local monotonicity in part (b)? Explain.
- Write down the demand function for good x at $p = p_0$ as a function of w and sketch the wealth-expansion path (for every $w > 0$). Note that you can sketch the wealth-expansion *without* solving for the demand, using your work from parts (a) and (b), and you should do that if you have trouble solving for the demand.

4. A variation of Robinson's economy has two goods, indexed by $\ell \in \{1, 2\}$. All firms are identical. Firm j uses the same technology to produce y_{j2} using y_{j1} where the second subindex represents the good:

$$Y_j = \{(y_{j1}, y_{j2}) : y_{j1} \leq 0, \quad y_{j2} \leq \max\{2, \max\{-(1 + y_{j1})^2 + 1, -(2 + y_{j1})^2 + 2\}\}\}$$

All consumers are identical. Consumer i has preferences $u_i(x_{i1}, x_{i2}) = \min\{x_{i1}, x_{i2}\}$, consumption set $X = \mathbb{R}_+^2$, the non-negative orthant, and endowment $\omega = (2, 0)$.

A carefully executed and explained graphical argument suffices (together with what calculations you find useful to execute your graphical argument).

- Suppose there is a single firm and a single consumer.
 - Find a Pareto optimal allocation.
 - Is there a Walrasian equilibrium? Either identify it or explain why it does not exist.
 - Can all Pareto optimal allocations become a Walrasian equilibrium with transfers?
- Suppose there are two identical firms, j and k , and two identical consumers. Is $p = (1, 1)$ a Walrasian equilibrium (with transfers) price vector? Please explain.